Reaction Kinetics Sheet A

1) a) For a reaction $A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B$:

$$t = \frac{1}{k_1 + k_{-1}} \ln\left(\frac{k_1[A]_0}{k_1[A]_0 - (k_1 + k_{-1})[B]}\right)$$

Show that this can be rearranged into the form:

$$[B] = \frac{[A]_0}{(k_{+1} + k_{-1})}(k_{+1} - k_{+1}\exp\{-(k_{+1} + k_{-1})t\})$$

$$t(k_1 + k_{-1}) = \ln\left(\frac{k_1[A]_0}{k_1[A]_0 - (k_1 + k_{-1})[B]}\right)$$

$$\exp[(k_1 + k_{-1})t] = \frac{k_1[A]_0}{k_1[A]_0 - (k_1 + k_{-1})[B]}$$

$$k_1[A]_0 - (k_1 + k_{-1})[B] = k_1[A]_0 \exp[-(k_1 + k_{-1})t]$$

$$-(k_1 + k_{-1})[B] = [A]_0(k_1 + k_1 \exp[-(k_1 + k_{-1})t])$$

$$[B] = \frac{[A]_0}{k_1 + k_{-1}}(k_1 + k_1 \exp[-(k_1 + k_{-1})t])$$

b) **(b)** If $[A] = \frac{[A]_0}{(k_{+1} + k_{-1})}(k_{-1} + k_{+1}\exp\{-(k_{+1} + k_{-1})t\})$, starting from time $t = 0$, by what proportion will [A] have decreased after a delay that lasts the length of the relaxation time, $\tau = 1/(k_{+1} + k_{-1})$?

Substitute $t = \tau = 1/k_1 + k_{-1}$

$$[A] = \frac{[A]_0}{k_1 + k_{-1}}(k_{-1} + k_1 \exp[-1])$$

$$\frac{[A]}{[A]_0} = \frac{k_{-1} + k_1/e}{k_{-1} + k_1}$$

(2) Following on from Equation (6.4) in the lecture handout, use partial fractions to show that:

$$\frac{1}{(p-[AB])(q-[AB])} = \frac{1}{(q-p)}\left(\frac{1}{(p-[AB])} - \frac{1}{(q-[AB])}\right)$$

$$\frac{1}{(p-[AB])(q-[AB])} = \frac{\alpha}{p-[AB]} + \frac{\beta}{q-[AB]}$$

$$\Rightarrow \quad 1 = \alpha(q-[AB]) + \beta(p-[AB]) = q\alpha - \alpha[AB] + \beta p - \beta[AB]$$

$$\Rightarrow \quad 1 = [AB](-\alpha-\beta) + q\alpha + p\beta$$

Compare coefficients $\quad -\alpha-\beta=0 \Rightarrow \alpha=-\beta$

$\therefore \quad q\alpha + p\beta = 1 \Rightarrow q\alpha - p\alpha = 1 \Rightarrow \alpha = \frac{1}{q-p}$

$$\therefore \quad \frac{1}{(p-[AB])(q-[AB])} = \frac{1/(q-p)}{p-[AB]} - \frac{1/(q-p)}{q-[AB]}$$

$$= \frac{1}{q-p}\left(\frac{1}{p-[AB]} - \frac{1}{q-[AB]}\right)$$

(3) In animal tissues, proteins are synthesized using only one optically active amino acid form, the *L*-enantiomer. Over time, however, *L*-amino acids slowly isomerise, producing measurable quantities of the *D*-enantiomer, via reactions that are not catalyzed by enzymes. If the rate constant for the isomerisation reaction is known, the age of an animal can be estimated using the ratio of the *D*- and *L*-forms of aspartic acid in samples of dentine taken from teeth.

(a) If the isomerisation reaction between the *L* and *D* enantiomers of aspartic acid obeys the chemical equilibrium equation:

$$L \underset{k}{\overset{k}{\rightleftharpoons}} D,$$ where the rate constants for the forward and backward directions are identical,

explain the significance of the following pair of coupled differential equations:

$$\frac{d[L]}{dt} = -k([L]-[D]) \quad \text{and} \quad \frac{d[D]}{dt} = +k([L]-[D])$$

$$\frac{d[L]}{dt} = k[D] - k[L] \qquad \frac{d[D]}{dt} = k[L] - k[D]$$

$D \xrightarrow{k} L \qquad L \xrightarrow{k} D$

(b) Determine a molar balance equation for the current concentrations of L and D in terms of their initial values.

Initial values @ $t=0$, $[L]_0 = [L]_0$, $[D]_0 = 0$

$$\therefore \quad [L]_0 = [L] + [D] \Rightarrow [L] = [L]_0 - [D]$$
$$\Rightarrow [D] = [L]_0 - [L]$$

(c) Show that t, the age of the sample, is given by:

$$t = -\frac{1}{2k} \ln\left[\frac{1 - [D]/[L]}{1 + [D]/[L]}\right]$$

Substituting from molar balance:

$$\frac{d[L]}{dt} = -k([L] - [L]_0 + [L]) = -k(2[L] - [L]_0)$$

$$\int_{[L]_0}^{[L]} \frac{d[L]}{2[L] - [L]_0} = \int_0^t -k\,dt = \left[\frac{1}{2}\ln(2[L] - [L]_0)\right]_{[L]_0}^{[L]} = -kt$$

$$\Rightarrow \frac{1}{2}\ln(2[L] - [L]_0) - \frac{1}{2}\ln([L]_0) = -kt$$

$$= \frac{1}{2}\ln\left(\frac{2[L] - [L]_0}{[L]_0}\right) = -kt$$

$$\Rightarrow t = \frac{-1}{2k}\ln\left(\frac{2[L] - [L]_0}{[L]_0}\right)$$

$$\Rightarrow t = \frac{-1}{2k}\ln\left(\frac{2[L] - [L] - [D]}{[L] + [D]}\right)$$

$$= \frac{-1}{2k}\ln\left(\frac{[L] - [D]}{[L] + [D]}\right) \quad \text{divide by } [L] \Rightarrow \frac{-1}{2k}\ln\left(\frac{1 - [D]/[L]}{1 + [D]/[L]}\right)$$

(d) Analysis of dentine from teeth of known age indicates that $k = 0.0020\ y^{-1}$. If the [D]/[L] ratio of a new sample is found to be 0.10, estimate its age.

$$t = \frac{-1}{0.0020\,y^{-1}} \ln\left(\frac{1-0.1}{1+0.1}\right) = 50.2 \text{ years} \approx 50 \text{ years.}$$

Sheet B